

Table Patients:

-- Insert Bills

75

■

INSERT INTO Bills VALUES

76

(501, 101, 1200.50, 'Paid', '2024-10-03'),

77

(502, 102, 850.00, 'Unpaid', '2024-10-06'),

78

(503, 103, 450.00, 'Paid', '2024-10-09'),

79

(504, 104, 300.00, 'Paid', '2024-10-13'),

80

(505, 105, 600.00, 'Unpaid', '2024-10-16'),

81

(506, 106, 1000.00, 'Paid', '2024-10-19'),

82

(507, 107, 750.00, 'Unpaid', '2024-10-23');

83

84

■

select * from Patients;

85

■

select * from Doctors;

86

■

select * from Appointments;

87

■

select * from Bills;

88

89

90

91

92

100%

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24:84

Result Grid





Filter Rows:

Q

Search

Edit:







Export/Import:





	patient_id	name	age	gender	city	doctor_id	
	101	Alice Johnson	45	Female	New York	1	
	102	Bob Carter	52	Male	Boston	2	
	103	Carol White	28	Female	Chicago	3	
	104	David King	35	Male	New York	1	
	105	Eva Green	12	Female	New York	5	
	106	Frank Miller	60	Male	Chicago	3	
	107	Grace Young	40	Female	Boston	2	
	NULL	NULL	NULL	NULL	NULL	NULL	

Table Doctors:

```
68 (1003, 103, 3, '2024-10-08', 'Knee Injury'),
69 (1004, 104, 1, '2024-10-12', 'Routine Check-up'),
70 (1005, 105, 5, '2024-10-15', 'Fever'),
71 (1006, 106, 3, '2024-10-18', 'Fracture Follow-up'),
72 (1007, 107, 2, '2024-10-22', 'Dizziness');
73
74 -- Insert Bills
75 ■ INSERT INTO Bills VALUES
76 (501, 101, 1200.50, 'Paid', '2024-10-03'),
77 (502, 102, 850.00, 'Unpaid', '2024-10-06'),
78 (503, 103, 450.00, 'Paid', '2024-10-09'),
79 (504, 104, 300.00, 'Paid', '2024-10-13'),
80 (505, 105, 600.00, 'Unpaid', '2024-10-16'),
81 (506, 106, 1000.00, 'Paid', '2024-10-19'),
82 (507, 107, 750.00, 'Unpaid', '2024-10-23');
83
84 ■ select * from Patients;
85 ■ select * from Doctors;
86 ■ select * from Appointments;
```

100% 23:85

Result Grid Filter Rows: Search Edit: Export/Import:

	doctor_id	name	specialization	experience_yea...	city	
	1	Dr. Smith	Cardiology	15	New York	
	2	Dr. Adams	Neurology	10	Boston	
	3	Dr. Patel	Orthopedics	12	Chicago	
	4	Dr. Lee	Dermatology	8	San Francisco	
	5	Dr. Brown	Pediatrics	5	New York	
	NULL	NULL	NULL	NULL	NULL	

Table Appointments:

75

■

INSERT INTO Bills VALUES

76

(501, 101, 1200.50, 'Paid', '2024-10-03'),

77

(502, 102, 850.00, 'Unpaid', '2024-10-06'),

78

(503, 103, 450.00, 'Paid', '2024-10-09'),

79

(504, 104, 300.00, 'Paid', '2024-10-13'),

80

(505, 105, 600.00, 'Unpaid', '2024-10-16'),

81

(506, 106, 1000.00, 'Paid', '2024-10-19'),

82

(507, 107, 750.00, 'Unpaid', '2024-10-23');

83

84

■

select * from Patients;

85

■

select * from Doctors;

86

■

select * from Appointments;

87

■

select * from Bills;

88

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100%

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28:86

Result Grid

 

Filter Rows:

Edit:

  

Export/Import:

 

	appointment...	patient_id	doctor_id	appointment_d...	diagnosis	
	1001	101	1	2024-10-02	Chest Pain	
	1002	102	2	2024-10-05	Migraine	
	1003	103	3	2024-10-08	Knee Injury	
	1004	104	1	2024-10-12	Routine Check-up	
	1005	105	5	2024-10-15	Fever	
	1006	106	3	2024-10-18	Fracture Follow-up	
	1007	107	2	2024-10-22	Dizziness	
	NULL	NULL	NULL	NULL	NULL	

Table Bills:

```

70  (1005, 105, 5, '2024-10-15', 'Fever'),
71  (1006, 106, 3, '2024-10-18', 'Fracture Follow-up'),
72  (1007, 107, 2, '2024-10-22', 'Dizziness');
73
74  -- Insert Bills
75  • INSERT INTO Bills VALUES
76  (501, 101, 1200.50, 'Paid', '2024-10-03'),
77  (502, 102, 850.00, 'Unpaid', '2024-10-06'),
78  (503, 103, 450.00, 'Paid', '2024-10-09'),
79  (504, 104, 300.00, 'Paid', '2024-10-13'),
80  (505, 105, 600.00, 'Unpaid', '2024-10-16'),
81  (506, 106, 1000.00, 'Paid', '2024-10-19'),
82  (507, 107, 750.00, 'Unpaid', '2024-10-23');
83
84  • select * from Patients;
85  • select * from Doctors;
86  • select * from Appointments;
87  • select * from Bills;
88
100% 21:87
Result Grid Filter Rows: Search Edit: Export/Import:


|  | bill_id | patient_id | total_amou... | payment_stat... | bill_date  |
|--|---------|------------|---------------|-----------------|------------|
|  | 501     | 101        | 1200.50       | Paid            | 2024-10-03 |
|  | 502     | 102        | 850.00        | Unpaid          | 2024-10-06 |
|  | 503     | 103        | 450.00        | Paid            | 2024-10-09 |
|  | 504     | 104        | 300.00        | Paid            | 2024-10-13 |
|  | 505     | 105        | 600.00        | Unpaid          | 2024-10-16 |
|  | 506     | 106        | 1000.00       | Paid            | 2024-10-19 |
|  | 507     | 107        | 750.00        | Unpaid          | 2024-10-23 |
|  | NULL    | NULL       | NULL          | NULL            | NULL       |


```

Part 1 – INNER JOINS

Q1.

Display each patient's name, doctor's name, and specialization.

Show patient's name ,doctor's name , doctors specialization

-- q1

```

select p.name as patient_name, d.name as doctor_name, d.specialization
from patients p
inner join doctors d on p.doctor_id = d.doctor_id;

```

```
71      (1006, 106, 3, '2024-10-18', 'Fracture Follow-up'),
72      (1007, 107, 2, '2024-10-22', 'Dizziness');
73
74      -- Insert Bills
75      INSERT INTO Bills VALUES
76      (501, 101, 1200.50, 'Paid', '2024-10-03'),
77      (502, 102, 850.00, 'Unpaid', '2024-10-06'),
78      (503, 103, 450.00, 'Paid', '2024-10-09'),
79      (504, 104, 300.00, 'Paid', '2024-10-13'),
80      (505, 105, 600.00, 'Unpaid', '2024-10-16'),
81      (506, 106, 1000.00, 'Paid', '2024-10-19'),
82      (507, 107, 750.00, 'Unpaid', '2024-10-23');
83
84      select * from Patients;
85      select * from Doctors;
86      select * from Appointments;
87      select * from Bills;
88
89      -- q1
```

100% 51:92

Result Grid Filter Rows: Search Export:

	patient_name	doctor_name	specialization	
	Alice Johnson	Dr. Smith	Cardiology	
	David King	Dr. Smith	Cardiology	
	Bob Carter	Dr. Adams	Neurology	
	Grace Young	Dr. Adams	Neurology	
	Carol White	Dr. Patel	Orthopedics	
	Frank Miller	Dr. Patel	Orthopedics	
	Eva Green	Dr. Brown	Pediatrics	

Q2.

List all appointment details along with doctor and patient names.

Show appointment_id, appointment_date, patient_name, doctor_name, diagnosis

-- q2

select a.appointment_id, a.appointment_date, p.name as patient_name, d.name as
doctor_name, a.diagnosis

from appointments a

inner join patients p on a.patient_id = p.patient_id

inner join doctors d on a.doctor_id = d.doctor_id;

```

87 select * from Bills;
88
89 -- q1
90 select p.name as patient_name, d.name as doctor_name, d.specialization
91 from patients p
92 inner join doctors d on p.doctor_id = d.doctor_id;
93
94 -- q2
95 select a.appointment_id, a.appointment_date, p.name as patient_name, d.name as doctor_name,
96 from appointments a
97 inner join patients p on a.patient_id = p.patient_id
98 inner join doctors d on a.doctor_id = d.doctor_id;
99
100
101
102
103
104

```

100% 51:98

Result Grid Filter Rows: Search Export:

	appointment...	appointment_d...	patient_name	doctor_name	diagnosis	
	1001	2024-10-02	Alice Johnson	Dr. Smith	Chest Pain	
	1004	2024-10-12	David King	Dr. Smith	Routine Check-up	
	1002	2024-10-05	Bob Carter	Dr. Adams	Migraine	
	1007	2024-10-22	Grace Young	Dr. Adams	Dizziness	
	1003	2024-10-08	Carol White	Dr. Patel	Knee Injury	
	1006	2024-10-18	Frank Miller	Dr. Patel	Fracture Follow-up	
	1005	2024-10-15	Eva Green	Dr. Brown	Fever	

Part 2 – LEFT & RIGHT JOINS

Q3.

Show all doctors and any patients assigned to them (if none, show NULL).

Show doctor_name, doctor's specialization, patient's_name

-- q3

```

select d.name as doctor_name, d.specialization, p.name as patient_name
from doctors d
left join patients p on d.doctor_id = p.doctor_id;

```

```

93
94 -- q2
95 select a.appointment_id, a.appointment_date, p.name as patient_name, d.name as doctor_name,
96 from appointments a
97 inner join patients p on a.patient_id = p.patient_id
98 inner join doctors d on a.doctor_id = d.doctor_id;
99
100 -- q3
101 select d.name as doctor_name, d.specialization, p.name as patient_name
102 from doctors d
103 left join patients p on d.doctor_id = p.doctor_id;
104
105
106
107
108
109
110

```

100% 51:103

Result Grid Filter Rows: Search Export:

	doctor_name	specialization	patient_name
	Dr. Smith	Cardiology	Alice Johnson
	Dr. Smith	Cardiology	David King
	Dr. Adams	Neurology	Bob Carter
	Dr. Adams	Neurology	Grace Young
	Dr. Patel	Orthopedics	Carol White
	Dr. Patel	Orthopedics	Frank Miller
	Dr. Lee	Dermatology	HULL
	Dr. Brown	Pediatrics	Eva Green

Q4.

List all patients (even those without bills) with their bill amounts and status.

Show patient_name, total_amount, payment_status

-- q4

```

select p.name as patient_name, b.total_amount, b.payment_status
from patients p
left join bills b on p.patient_id = b.patient_id;

```

```
98     inner join doctors d on a.doctor_id = d.doctor_id;
99
100    -- q3
101    select d.name as doctor_name, d.specialization, p.name as patient_name
102    from doctors d
103    left join patients p on d.doctor_id = p.doctor_id;
104
105    -- q4
106    select p.name as patient_name, b.total_amount, b.payment_status
107    from patients p
108    left join bills b on p.patient_id = b.patient_id;
109
110
111
112
113
114
115
```

100% 50:108

Result Grid Filter Rows: Search Export:

	patient_name	total_amou...	payment_stat...	
	Alice Johnson	1200.50	Paid	
	Bob Carter	850.00	Unpaid	
	Carol White	450.00	Paid	
	David King	300.00	Paid	
	Eva Green	600.00	Unpaid	
	Frank Miller	1000.00	Paid	
	Grace Young	750.00	Unpaid	

Q5.

Display all bills and, if available, show the patient's city and doctor.

Show bill_id, total_amount, payment_status, patient_name, p.city, doctor_name

-- q5

```
select b.bill_id, b.total_amount, b.payment_status, p.name as patient_name, p.city, d.name
from bills b
left join patients p on b.patient_id = p.patient_id
left join doctors d on p.doctor_id = d.doctor_id;
```



```

102     from doctors d
103     left join patients p on d.doctor_id = p.doctor_id;
104
105     -- q4
106     select p.name as patient_name, b.total_amount, b.payment_status
107     from patients p
108     left join bills b on p.patient_id = b.patient_id;
109
110     -- q5
111     select b.bill_id, b.total_amount, b.payment_status, p.name as patient_name, p.city, d.name as doctor_name
112     from bills b
113     left join patients p on b.patient_id = p.patient_id
114     left join doctors d on p.doctor_id = d.doctor_id;
115
116
117
118
119

```

100% 1:115

Result Grid Filter Rows: Search Export:

	bill_id	total_amo...	payment_stat...	patient_name	city	doctor_name
	501	1200.50	Paid	Alice Johnson	New York	Dr. Smith
	502	850.00	Unpaid	Bob Carter	Boston	Dr. Adams
	503	450.00	Paid	Carol White	Chicago	Dr. Patel
	504	300.00	Paid	David King	New York	Dr. Smith
	505	600.00	Unpaid	Eva Green	New York	Dr. Brown
	506	1000.00	Paid	Frank Miller	Chicago	Dr. Patel
	507	750.00	Unpaid	Grace Young	Boston	Dr. Adams

Part 3 – MULTIPLE JOINS & FILTERS

Q6.

List the doctor's name, patient's name, diagnosis, and bill amount for only paid bills.

Show doctor_name, patient_name, diagnosis, total_amount

-- q6

```

select d.name as doctor_name, p.name as patient_name, a.diagnosis, b.total_amount
from bills b
inner join patients p on b.patient_id = p.patient_id
inner join doctors d on p.doctor_id = d.doctor_id

```

inner join appointments a on p.patient_id = a.patient_id
where b.payment_status = 'Paid';

```
109
110 -- q5
111 • select b.bill_id, b.total_amount, b.payment_status, p.name as patient_name, p.city, d.name a
112   from bills b
113   left join patients p on b.patient_id = p.patient_id
114   left join doctors d on p.doctor_id = d.doctor_id;
115
116 -- q6
117 • select d.name as doctor_name, p.name as patient_name, a.diagnosis, b.total_amount
118   from bills b
119   inner join patients p on b.patient_id = p.patient_id
120   inner join doctors d on p.doctor_id = d.doctor_id
121   inner join appointments a on p.patient_id = a.patient_id
122   where b.payment_status = 'Paid';
123
124
125
126
```

100% 33:122

Result Grid Filter Rows: Search Export:

	doctor_name	patient_name	diagnosis	total_amou...	
	Dr. Smith	Alice Johnson	Chest Pain	1200.50	
	Dr. Patel	Carol White	Knee Injury	450.00	
	Dr. Smith	David King	Routine Check-up	300.00	
	Dr. Patel	Frank Miller	Fracture Follow-up	1000.00	

Q7.

Find all appointments handled by doctors from
New York.

Show doctor_name, patient_name, appointment_date, diagnosis

```
114     left join doctors d on p.doctor_id = d.doctor_id;
115
116     -- q6
117     select d.name as doctor_name, p.name as patient_name, a.diagnosis, b.total_amount
118     from bills b
119     inner join patients p on b.patient_id = p.patient_id
120     inner join doctors d on p.doctor_id = d.doctor_id
121     inner join appointments a on p.patient_id = a.patient_id
122     where b.payment_status = 'Paid';
123
124     -- q7
125     select d.name as doctor_name, p.name as patient_name, a.appointment_date, a.diagnosis
126     from appointments a
127     inner join doctors d on a.doctor_id = d.doctor_id
128     inner join patients p on a.patient_id = p.patient_id
129     where d.city = 'New York';
130
131
```

100% 27:129

Result Grid Filter Rows: Search Export:

	doctor_name	patient_name	appointment_d...	diagnosis	
	Dr. Smith	Alice Johnson	2024-10-02	Chest Pain	
	Dr. Smith	David King	2024-10-12	Routine Check-up	
	Dr. Brown	Eva Green	2024-10-15	Fever	

```
-- q8
select p.patient_id, p.name as patient_name, b.bill_id, b.total_amount, b.payment_status
from patients p
left join bills b on p.patient_id = b.patient_id
union
select p.patient_id, p.name as patient_name, b.bill_id, b.total_amount, b.payment_status
from patients p
right join bills b on p.patient_id = b.patient_id;
```

```

123
124 -- q7
125 select d.name as doctor_name, p.name as patient_name, a.appointment_date, a.diagnosis
126 from appointments a
127 inner join doctors d on a.doctor_id = d.doctor_id
128 inner join patients p on a.patient_id = p.patient_id
129 where d.city = 'New York';
130
131 -- q8
132 select p.patient_id, p.name as patient_name, b.bill_id, b.total_amount, b.payment_status
133 from patients p
134 left join bills b on p.patient_id = b.patient_id
135 union
136 select p.patient_id, p.name as patient_name, b.bill_id, b.total_amount, b.payment_status
137 from patients p
138 right join bills b on p.patient_id = b.patient_id;
139
140

```

100% 51:138

Result Grid Filter Rows: Search Export:

	patient_id	patient_name	bill_id	total_amou...	payment_stat...
	101	Alice Johnson	501	1200.50	Paid
	102	Bob Carter	502	850.00	Unpaid
	103	Carol White	503	450.00	Paid
	104	David King	504	300.00	Paid
	105	Eva Green	505	600.00	Unpaid
	106	Frank Miller	506	1000.00	Paid
	107	Grace Young	507	750.00	Unpaid

Q9.

Find doctors who work in the same city as
Dr. Smith.

Show doctor name and City

```

select d2.name, d2.city
from doctors d1
inner join doctors d2 on d1.city = d2.city
where d1.name = 'Dr. Smith' and d2.name <> 'Dr. Smith';

```

```

130
131 -- q8
132 select p.patient_id, p.name as patient_name, b.bill_id, b.total_amount, b.payment_status
133 from patients p
134 left join bills b on p.patient_id = b.patient_id
135 union
136 select p.patient_id, p.name as patient_name, b.bill_id, b.total_amount, b.payment_status
137 from patients p
138 right join bills b on p.patient_id = b.patient_id;
139
140 -- q9
141 select d2.name, d2.city
142 from doctors d1
143 inner join doctors d2 on d1.city = d2.city
144 where d1.name = 'Dr. Smith' and d2.name <> 'Dr. Smith';
145
146
147

```

100% 56:144

Result Grid



Filter Rows:

Search

Export:



	name	city	
	Dr. Brown	New York	

-- q10

```

select d.name as doctor_name, sum(b.total_amount) as total_collected
from bills b
inner join patients p on b.patient_id = p.patient_id
inner join doctors d on p.doctor_id = d.doctor_id
where b.payment_status = 'Paid'
group by d.name;

```

```

138     right join bills b on p.patient_id = b.patient_id;
139
140     -- q9
141     select d2.name, d2.city
142     from doctors d1
143     inner join doctors d2 on d1.city = d2.city
144     where d1.name = 'Dr. Smith' and d2.name <> 'Dr. Smith';
145
146     -- q10
147     select d.name as doctor_name, sum(b.total_amount) as total_collected
148     from bills b
149     inner join patients p on b.patient_id = p.patient_id
150     inner join doctors d on p.doctor_id = d.doctor_id
151     where b.payment_status = 'Paid'
152     group by d.name;
153
154
155

```

100% 17:152

Result Grid



Filter Rows:



Search

Export:



	doctor_name	total_collect...	
	Dr. Smith	1500.50	
	Dr. Patel	1450.00	